

eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus

Alday, John Denrick M., Canuel, Mark Justin R., Eusebio, Rain Justine R., Familara, Carl Justin B., Gaoat, Athena Jem D., Huerto, Benlester N., Laguitao, Maria Angela B., Parale, Carl Kristian A., Singian, Em-Jay P., Verdan, Geryk Drew P.

**ICCT Colleges
Bachelor of Science in Information Technology (BSIT)**

ABSTRACT

Students at ICCT Colleges frequently encounter challenges with fragmented academic information, repetitive inquiries, and information uncertainty regarding school documents and procedures. They currently rely on multiple disconnected channels such as Facebook, bulletin boards, and word-of-mouth, leading to confusion and incomplete submissions. To address these issues, this study developed eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus — a platform designed to centralize fragmented requirement information, provide clear step-by-step guidance, and deliver accurate and reliable information to reduce student uncertainty. The system features a centralized documents guide, step-by-step procedural instructions, admin-posted announcements, email notifications, and a chatbot for immediate assistance. This study contributes to the improvement of student and administrative services at ICCT Colleges by providing a digital framework for streamlined information dissemination, expected to reduce the time students spend searching for document procedures, minimize repetitive office visits, and offer a reliable source of academic information. This study utilized a Design Science Research (DSR) approach, followed by the Waterfall Software Development Life Cycle, and evaluated the system using a mixed methods approach, combining quantitative data from 160 student respondents through Likert-scale surveys and User Acceptance Testing (UAT) with qualitative feedback from administrative staff interviews. Results demonstrated that the system successfully achieved all four objectives, with overall weighted means ranging from 4.28 to 4.45 and a UAT functional test pass rate of 99.57%.

KEYWORDS

web-based application, guidance system, school documents, information fragmentation, repetitive inquiries, information uncertainty, centralized platform, mixed methods, user acceptance testing

CHAPTER 1

INTRODUCTION

A. Background and Motivation

In this period of time, colleges and universities use systems to help students and school administrators complete their respective tasks efficiently [1], [3], [12]. Institutions that

remain dependent on outdated, manual methods such as face-to-face inquiries and paper-based information create a significant technological and operational gap. At ICCT Colleges, these gaps become evident as information is fragmented across different platforms, creating confusion and forcing students to ask the same questions repeatedly [4]. A survey with 100 students confirmed these issues: 71% reported needing to check multiple platforms to find document requirement information, 73% admitted asking the same question repeatedly due to unclear instructions, and 72% expressed feeling unsure about the accuracy of the information they found. This environment of ambiguity directly generates information fragmentation, repetitive inquiries, and information uncertainty among students. This context demonstrates that education should proceed by supporting digital solutions, web-based systems, and software platforms for managing school documents, requirements, information, and procedures [5], [9], [12].

This capstone is motivated by the demonstrated gap between ICCT Colleges' current operational model and evidence-based technological solutions proven to improve academic administrations [8]. A Web-Based School Document Guide System follows this basis, and it focuses on providing ICCT Colleges students with an understandable and easily available system for accessing school document procedures and school announcements. Hence, this system will help lessen the confusion among all students. The system also further enhances communication and understanding between students and school staff, encouraging efficient procedures and responsibility in accomplishing the documents.

B. Problem Statement

General Problem

Educational institutions that rely on disconnected and manual methods to disseminate school documents and requirements procedures and information create significant

problems for students. Important information gets lost or becomes vague, leaving students confused and unsure of what to follow. As a result this hinders their academic progress and overall success.

Specific Problem

This study addresses three specific, interlinked problems at ICCT Colleges :

1. **Information Fragmentation:** School requirement information is scattered across Facebook, bulletin boards, and word-of-mouth. Students cannot find all document procedures in one place, and posts get buried or lost, creating confusion.
2. **Repetitive Inquiries:** The lack of clear, step-by-step instructions forces students to ask the same questions multiple times, visit offices repeatedly, and submit incomplete documents and requirements.
3. **Information Uncertainty:** Students cannot verify if requirement information is accurate or official. Conflicting information from Facebook and other students leads to inconsistent submissions and repeated requests for staff confirmation.

C. Objectives of the Study

General Objective

The general objective of this study is to design, develop, and deploy eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus providing step-by-step guidance and school announcements. This will handle academic inquiries efficiently and provide ICCT Colleges students with a centralized digital platform for accessing school documents and requirements with step-by-step guidance and school announcements.

Specific Objectives

1. To centralize fragmented requirement information from multiple sources (Facebook, bulletin boards, word-of-mouth) into a single, organized digital platform.
2. To provide clear, step-by-step guidance that enables students to complete document requirements without repeated office visits.
3. To ensure accurate, reliable, and consistent information that reduces student confusion and uncertainty.
4. To evaluate the usability and effectiveness of the system at ICCT Main Campus.

D. Scope and Limitations

This study focuses on the creation and evaluation of eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus.

The system is designed for students and administrative staff of ICCT Colleges. To make it convenient and widely accessible, it will be available through web browsers on both computers and mobile devices. The main purpose of this system is to reduce confusion about school document requirements, processes, and relevant information; it will also minimize repetitive questions and improve the communication of the school community.

It is designed to be a support tool that makes important school information easier to access. The notifications system will not provide real-time messages, but instead it will send email notifications only when an administrator posts a new announcement. The system is designed as an informational and procedural guide. Also, it will not replace the official announcements or policies of the school.

Scope of the Study

- **Main function:** To serve as a support tool that makes school requirement procedure information easier to access.
- **Primary users:** Students of ICCT Colleges Cainta are the primary end-users.
- **Secondary users:** Administrative staff are secondary users.
- **Platform:** A web application that students and administrative staff can access through browsers on computers and mobile devices.
- **Location:** The system is designed with the intent to serve all ICCT Students. However, for the purpose of this capstone and development project, the ICCT Main Campus will serve as one and only pilot site. This ensures that the project scope remains feasible while introducing a foundational framework that could be adapted for other campuses.

Limitations of the Study

- **Notification type:** it will send a notification through email and will not provide a real-time message or live alerts.
- **System role:** It is an informational and procedural guide about school document requirements, and school announcement not a transactional or policy platform.
- **Chatbot:** The chatbot does not retain chat history. Each interaction starts fresh, without context from previous conversations.

E. Significance of the Study

The researchers conducted this study to provide an efficient and accessible guide for the students to follow for their school document requirements. This system seeks to simplify school processes, promote time management, and

improve communication between students and the school department. The following are the ones who will benefit from this research:

Students. For students, this study will greatly benefit them by giving a centralized platform for accessing step-by-step procedures and reminders for their school document requirements. This study will also reduce the time spent by all the students who are seeking the step-by-step procedure and school requirements.

School Administrators. For School Administrators, this study will benefit them by giving them access to update or add school document procedures and announcements effortlessly. This study will also make the dissemination of updated or new school document requirements and announcements easily accessible to all students.

ICCT Colleges. For the institutions, this study will benefit them by promoting digital transformation within the school. This study will also enhance the institution's service quality and support a more student-centered academic environment.

Future Researchers. For future researchers, this study provides a documented case of applying Design Science Research and Waterfall methodology to develop a centralized academic guide system. The documented challenges particularly in content gathering, user authentication design, and notification implementation can serve as practical references for similar system development projects in higher education institutions.

CHAPTER 2 RELATED WORK

A. Review of Relevant Literature

The COVID-19 pandemic highlighted the growing importance of digital technologies in higher education for supporting teaching, learning, and student engagement. Nkomo, Daniel, and Butson [1] found that web portals and centralized platforms help students access academic information more efficiently and promote greater interaction with learning materials. Likewise, the eGuide ICCT system was developed to address fragmented academic information and communication delays by providing a centralized, web-based platform for timely updates and organized access to school document requirements procedure.

The role of computer-based technologies in enhancing student engagement has been widely recognized in higher education. Schindler et al. [2] found that digital tools positively influence students' behavioral, emotional, and cognitive engagement, and similarly, eGuide ICCT applies computer-based technology through a centralized web-based system for school document requirements procedure and announcements. Both emphasize how digital solutions improve communication, engagement, and overall efficiency in academic settings. In addition, Digital technologies play a crucial role in improving learning and information management in higher education. Pinto and Leite [3] digital

platforms enhance student engagement, communication, and access to academic materials, while the eGuide ICCT system developed for ICCT Colleges applies a web-based approach to provide students with organized school document requirements procedure and announcements, making sure students know exactly what to do instead of just guessing based on unreliable rumors.

Fragmented digital information significantly affects how academic institutions manage and deliver essential school-related content. H. Han Xu [4] emphasized that scattered online information in digital learning environments can lead to confusion, superficial understanding, and inefficient communication among students and educators. Similarly, the proposed eGuide ICCT system addresses the issue of fragmented academic information by providing a centralized web-based platform for school document requirements procedure and announcements. By organizing and streamlining information dissemination. Similarly, the web portals in higher education have been shown to improve information management and streamline communication. Pinho, Franco, and Mendes [5] emphasized that centralized platforms help students, teachers, and administrators access critical information efficiently. This concept supports eGuide ICCT, a web-based system designed to organize school document requirements procedures and provide timely announcements to ICCT students.

Recent research highlights how the web-based information system becomes a vital role in improving service delivery and strengthening communications in educational institutions. Zhiyong et al. [6] Explains that digital platforms can increase the user satisfaction and organizational effectiveness of information especially when it is clearly organized, and the processes are simplified. Likewise, the online academic systems are minimizing the scattered information by gathering announcements and document requirements procedure into one accessible platform, which makes it easier for users to follow the right procedures and meet the deadlines. By cutting down repeated questions and allowing students to independently access the reliable information, these systems will help reduce the workloads of administrations. In addition to that, just like our system, admin-posted announcements will keep students updated about school announcements and school announcements like events, which will prevent the students from becoming late or not knowing the events that will be held in our school. This idea supports the development of our system, which is designed to solve issues such as fragmented information, repetitive Inquiries and Information Uncertainty. Through a centralized platform the system will improve the communication flow, lowering the students anxiety, and have more efficient and structured academic experience for ICCT students.

M. Muso [7] explored the development of a web-based academic information system which is designed to

improve the institutional service and improve the access for the student's streamline processes. Their research found that centralized digital platforms help students easily reduce confusion, delays associated with manual inquiries, and obtain academic requirements. Also, it highlighted the integrating clear procedures and online access improves efficiently for both administrative staff and students. Furthermore the system demonstrated how the digital solutions ensure that students receive accurate and timely information and enhance communication flow. These findings support the objective of eGuide ICCT system, which it seeks to address the fragmented academic information and inefficient communication channels. Also, their system uses SMS and WhatsApp for notification. On the other hand, our system only focuses on using email for our announcement notification. eGuide ICCT provides an accessible and centralized document procedure guide for ICCT students to promote a more efficient and organized experience.

University students often have difficulty keeping track of assignments, deadlines, and course materials due to busy schedules and multiple learning platforms. To address this, AutoRemind, inspired by Education Support – Brightspace Manuals (Wageningen University & Research) [8] is designed as a centralized notification tool that delivers personalized reminders and integrates with Learning Management Systems to provide real-time updates on tasks. Likewise, eGuide ICCT provides a web-based platform that consolidates academic document requirements procedure and campus announcements, reducing confusion and improving access to essential information

The Educational IR Support Group of Kyushu Institute of Technology [9] explained that although the university already implemented an e-portfolio system to help students and teachers monitor learning progress and conduct self-evaluations, the collected data were stored across multiple systems and managed separately. This lack of integration limited the university's ability to fully utilize academic data for institutional research and decision-making, which was addressed by introducing an ETL tool and a centralized data lake system to consolidate and manage data more efficiently. Similarly, eGuide ICCT was developed to address challenges at ICCT Colleges, where students face scattered academic information, repetitive inquiries, and information uncertainty.

Higher Education Institutions often face challenges in managing multiple fragmented information systems that support academic, administrative, and strategic operations. Sequeira et al. [10] this problem is addressed by integrating systems using Business Intelligence (BI), Decision Support Systems (DSS), and AI-driven tools to enable data-driven decision-making, improve operational efficiency, and align technology with institutional goals. Likewise, the development of eGuide ICCT implements a web-based platform to organize school document requirements procedure and announcements, aiming to reduce confusion, improve

communication, and enhance accessibility of information for both students and administrative staff at ICCT Colleges.

Baglioni, Mannocci, Pavone, De Bonis, and Manghi [11] further emphasized that multiple registrations and fragmented information sources force users to juggle multiple platforms, leading to redundancy and confusion. This directly supports the problem of information fragmentation at ICCT Colleges, where students navigate Facebook, bulletin boards, and word-of-mouth to find document requirement information.

B. Review of Related Systems and Studies

Multiple studies have demonstrated the effectiveness of web-based academic systems. Caratquit [12] and Zamora Sr. [13] both reported that centralized platforms enable faster sharing of announcements and improve access to academic information. Similarly, Shelke, Khan, and Kapse [14] and Kizito and Nwabueze [15] found that student management systems reduce manual errors and streamline daily tasks. These findings support the development of eGuide ICCT, which applies a similar centralized approach to organize school document requirements and provide announcements to students.

The implementation of web-based academic systems facilitates more efficient handling of student and administrative tasks. Javelosa et al. [16] developed a CISAT enrollment system with features for scheduling, registration, profile management, and payment processing, allowing students, teachers, and administrators to perform tasks efficiently through role-specific modules. Applying this concept, eGuide ICCT organizes school document requirements procedure and campus announcements to reduce confusion and save time. Both systems improve accessibility, efficiency, and role-specific task management through online system automation. In comparison, Web-based school information systems play a crucial role in improving the accessibility and management of academic information in educational institutions. K. Wahab et al. [17] developed a Student Portal System that enabled faster sharing of announcements, assignments, and learning resources, allowing students and staff to stay informed and organized. In the same way, eGuide ICCT applies this approach by providing a centralized web-based platform for school document requirements procedure and announcements, reducing confusion and enhancing the communication of academic information. On the other hand, Centralized student information systems are essential for efficiently managing academic data and improving information dissemination within educational institutions. Sindhu et al. [18] reported that a web-based student information system tracks enrollment, attendance, grades, and faculty data, providing a centralized platform to streamline academic and administrative operations. Similarly, the development of eGuide ICCT implements a web-based system to organize school document

requirements procedure and aims to reduce confusion, improve communication, and enhance accessibility of information for both students and administrative staff at ICCT Colleges.

The use of web-based academic information systems improves access to educational services and communication between school offices and students S. Laugi [19]. The researcher found that students obtain academic information more efficiently, and reduce the confusion caused by manual processes and scattered announcements by the help of digital platforms. Also, it emphasizes the centralized systems improve transparency and ensure the students will receive accurate and consistent information about the academic procedures. These findings are comparable with the objective of our study, which addresses delayed communication, and fragmented information that is identified in the study. In addition, the use of organized digital content and online notifications was shown to enhance the awareness of students to deadlines and institutional updates. Both highlight how centralized web-based platforms can reduce the administrative burden, improve the communication flow, and create a more organized academic environment.

R. Aquino et al [20] examined the development of a web-based academic support or information management system, which is designed to improve quality communication, access to information, and the overall efficiency of the educational institution. Their system helped reduce the workload of the administrative staff of the school by minimizing the repetitive questions, inquiries, and step by step process of the important documents they needed. It is similar to our research, because we propose a system that emphasizes the importance of having a user-friendly interface with a clear step by step process of what is needed for a better user satisfaction. Both systems are aiming to address the issue of delayed communication and scattered information that makes the time of other people being wasted. Furthermore, the notification or announcement features of our system will keep the students updated to the dates and deadlines of important events, such as day of enrollment of your department. This supports the development of the eGuide ICCT, because this proves that this web-based guide system can improve transparency and enhance the overall institutional efficiency.

Sukma et al [21] from Politeknik Negeri Padang, Indonesia, the researchers developed an online platform for academic advising. Before they created a system, student consultations were done manually, which often led to scheduling problems and a lack of proper records or not accurate answers to their questions which makes it hard for students. To solve these issues, they designed a web-based guidance system using the Waterfall development method. This platform allows students and advisors to communicate online where they manage consultation logs, and monitor academic progress efficiently. The researchers found out that this type of system is more convenient and organized for all

students. Also, it is more accurate since department heads update the information if there is any wrong or changed information. It allows for faster information gathering of students and makes their questions answered real time. This study is connected to our study since we also want to create a system which helps the students with questions about information of document requirements or scholarships in our school. With the help of it, we can guide the students through the requirements and process of what they need to do.

Benguet State University [22] recently launched a student portal to provide students with accessible digital services. The university emphasized that the portal is "a bridge that connects learning, communication, and opportunity in a more efficient and student-centered way." This demonstrates that Philippine higher education institutions are increasingly adopting web-based platforms to improve student access to information, supporting the development of eGuide ICCT at ICCT Colleges.

C. Research Gap Analysis

A research gap is an area within a discipline that remains inadequately investigated. Among recognized categories, a contextual gap arises when studies fail to account for specific populations, environments, or geographical localities ResearchMethod.net [23]. This gap is directly relevant to the present study. While numerous academic institutions have adopted web-based platforms for information management, none have specifically targeted the integration of school document guidelines, step-by-step procedural instructions, and centralized announcements within the particular setting of ICCT Colleges.

At ICCT Colleges, students currently rely on fragmented sources such as Facebook, bulletin boards, and word-of-mouth to obtain school requirement procedures, leading to information fragmentation, repetitive inquiries, and information uncertainty [4], [11]. These challenges have not been systematically addressed by any existing web-based platform at ICCT Colleges, as current systems primarily focus on enrollment, attendance, or grade management rather than on providing a dedicated procedural guide.

By directing its focus toward this previously overlooked institutional environment, eGuide ICCT bridges this contextual gap. Through its centralized document guide, step-by-step instructions, admin-posted announcements, automated email notifications, and chatbot assistance, eGuide ICCT provides a tailored solution that directly addresses the distinct issues of information dispersal and communication inefficiencies at ICCT Cainta Main Campus.

CHAPTER 3 METHODOLOGY

A. Research Design

This study is guided by the Design Science Research (DSR), which focuses on creating and evaluating IT artifacts to solve real-world problems. DSR is appropriate since the primary output of this study is the eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus. The system is designed to address the three core problems identified at ICCT Colleges: information fragmentation, repetitive inquiries, and information uncertainty. This emphasizes practical relevance, rigorous evaluation, and contribution to both practice and knowledge [1]. This study executed the following phases:

Table 3.1: DSR Phases and Activities

Phase	Activity in this Study
1. Problem Identification	Identify information fragmentation, repetitive inquiries, and information uncertainty at ICCT Colleges through surveys.
2. Define Objectives	Define system requirements including centralization, school document requirements guide procedure, announcement notification (admin-posted only). .
3. Design and Development	Develop the eGuide ICCT using Waterfall methodology.
4. Demonstration	Deploy the system in pilot implementation at the ICCT Main Campus.
5. Evaluation	Assess the system usability, performance, and impact using Surveys and User Acceptance Testing (UAT).
6. Communication	Document the results in this capstone paper.

B. Development Methodology

The eGuide ICCT system was developed using the Waterfall methodology, which is aligned with the nature of Design Science Research (DSR). The development followed a sequential, phase-based approach, with each phase completed before proceeding to the next. The Waterfall methodology was selected for several reasons. First, the system requirements were well-defined from the start based on the pre-implementation surveys, which identified the three core problems to be addressed. Second, the project followed a fixed timeline typical of capstone research, making Waterfall's sequential phases suitable for planning and milestone tracking. Third, user feedback was collected primarily after full system

implementation through UAT and surveys, rather than during development. Given that the scope and objectives remained stable throughout the development process. This sequential approach also aligns with the Design Science Research framework, where the IT artifact is built and then evaluated in a structured manner.

Figure 3.1: Waterfall Software Development Life Cycle

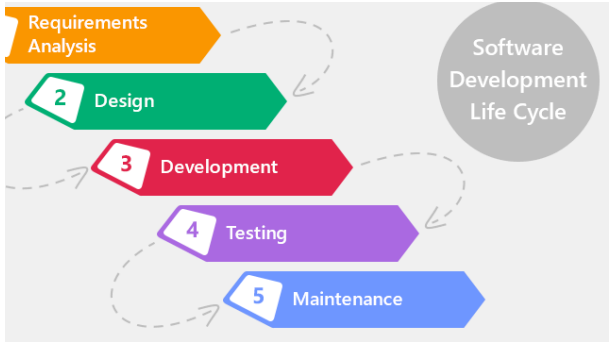


Table 3.2: Waterfall Development Phases and Outputs

Phase	Duration	Focus	Output
Phase 1	2 weeks	Requirements Analysis	Identified system requirements based on existing problems .
Phase 2	2 weeks	UI/UX and Frontend Design	Responsive web with desktop and mobile interface design
Phase 3	2 weeks	User Authentication and Database .	Login & Sign up system, RBAC, database schema
Phase 4	2 weeks	Centralized Information Module	Document procedure uploading, Announcement posting, search, step-by-step guides, search announcements
Phase 5	2 weeks	Notification System and Chatbot	Email notifications, chatbot functionality
Phase 6	2 weeks	Integration and Testing	System integration, bug fixes, UAT preparation

C. Research Approach

This study employs a mixed methods research approach, combining quantitative data from student surveys and UAT with qualitative feedback from administrative staff interviews.

Table 3.3: Research Approach: Quantitative and Qualitative Methods

Type	Purpose	Methods
Quantitative	Provides measurable results for evaluating users needs, problems, experience, and the systems usability, performance, and impact.	Likert scale survey, and User Acceptance Testing (UAT).
Qualitative	Provides in-depth insights and contextual feedback from administrative staff.	Semi-structured interviews, open-ended feedback.

D. Research Instrument

In this study, we designed our research instrument to capture quantitative and qualitative data. We employed measurable methods to evaluate system performance, usability, and impact, including the Likert-scale surveys and User Acceptance Testing (UAT), which allowed us to transform human feedback into statistical evidence. In addition to the quantitative instruments, qualitative feedback was gathered from administrative staff through open-ended discussions after system demonstration. This qualitative data was analyzed thematically to identify recurring suggestions and insights.

We distributed the instrument to our chosen respondents in an efficient and systematic manner. To effectively reach a larger sample size for our quantitative data, we distributed surveys using both physical handouts and digital platforms like Google Forms. Every respondent received clear instructions and an informed consent form from us, outlining the goal of the research and guaranteeing the privacy of their answers. This thoughtful and ethical distribution strategy allowed us to preserve the integrity of our research and maintain a high response rate.

E. Respondents

The primary respondents of this study are students from ICCT Colleges Main Campus. The efficiency and usefulness of the proposed eGuide ICCT system were determined using their answers. Supplementary feedback was gathered from administrative staff, but the formal evaluation focuses on student outcomes as stated in the study objectives.

Participation was voluntary, and confidentiality and data privacy were strictly upheld.

The respondents were selected using convenience sampling. As noted by Gall, Borg, and Gall (1996), as cited by the University of Connecticut [24], while this is not the strongest sampling procedure, it is often better to conduct a study with a convenience sample than to conduct no study at all, provided the sample suits the purpose of the study. The researchers targeted students who were available and willing to participate during the pilot deployment period [7]. All year levels were included to capture a diverse range of experiences with school document requirements. The demographic distribution (Table 4.1) reflects the actual student population at ICCT Cainta Main Campus, with third-year students comprising the largest group.

While the exact total student population of ICCT Cainta Main Campus was not available for this study, as the school does not have a confirmed count of students currently enrolled at the Main Campus, the sample size of 160 respondents is considered sufficient for statistical analysis. According to Memon et al. [25], who reviewed established guidelines for survey research, Roscoe's rule of thumb suggests that a sample size greater than 30 and less than 500 is suitable for most studies. Given the feasibility constraints of this capstone project, the sample of 160 is considered adequate for the purposes of this study.

F. Data Gathering Methods

The data gathering for eGuide ICCT will be conducted through a systematic process to ensure that the development of the centralized web-based system is grounded in real student needs. Since the study follows the Design Science Research (DSR) framework and a Waterfall methodology, data collection is divided into phases: from identifying the initial problem through surveys to the final evaluation of the system using post-implementation surveys and User Acceptance Testing (UAT).

1. Securing Permission and Ethics

Before approaching any participants, we secured formal approval from the ICCT Colleges administration by sending a letter informing them of our study. We ensured that our study was fully authorized. We communicated to respondents that their participation is entirely voluntary and that their identities will remain anonymous, keeping our data handling professional and ethical.

2. System Evaluation (Planned Quantitative Survey)

Once the system is fully developed and piloted, we will conduct our primary quantitative data gathering. This is

the stage where we will distribute a structured survey questionnaire. We plan to use a Likert Scale to measure:

- How easy the system is to use.
- If the announcement notifications from admins are helpful.
- If the step-by-step guides reduce student confusion.

We also intend to use User Acceptance Testing (UAT) to assess whether the system meets the functional needs of students. The UAT will be administered via Google Forms during pilot testing.

3. Data Organization and Analysis Plan

After collecting the responses, data will be organized for analysis. Likert-scale results from Sections 1-4 and UAT evaluations will be encoded into a structured data sheet. Descriptive statistics, specifically the weighted mean and standard deviation, will be calculated for each section to determine whether the eGuide ICCT system achieved the success criteria (mean ≥ 3.41 , interpreted as 'Agree'). Interpretation ranges were: 1.00-1.80 (Strongly Disagree), 1.81-2.60 (Disagree), 2.61-3.40 (Neutral), 3.41-4.20 (Agree), and 4.21-5.00 (Strongly Agree).

Functional testing (Pass/Fail) results from Section 5 will be analyzed using frequency distribution to calculate the percentage of 'Pass' responses for each test case. The overall UAT acceptance rate from Section 6 will be calculated as the percentage of respondents who selected 'ACCEPT'.

G. Data Analysis Plan

This section outlines the systematic approach for analyzing the data collected to evaluate the effectiveness of the proposed eGuide ICCT system at ICCT Colleges. This study applied a mixed methods approach to address the problems identified in Chapter 1: information fragmentation, repetitive inquiries, and information uncertainty. The quantitative data provide measurable insights into student perceptions and system functionality, while the qualitative feedback from administrative staff provides contextual insights and suggestions for improvement.

Qualitative Data Analysis:

Qualitative feedback gathered from administrative staff through open-ended discussions was analyzed using thematic analysis. The responses were reviewed, coded, and organized into recurring themes such as ease of navigation, login method suggestions, account recovery, and security concerns. Key quotes were extracted to provide context and support for the quantitative findings. This qualitative analysis complements the quantitative results by offering deeper insights into the system's usability and areas for improvement.

Data Preparation and Cleaning:

Before the analysis begins, all collected data will undergo preparation and cleaning procedures to ensure integrity and accuracy.

Quantitative Data:

- Incomplete survey responses will be reviewed. Surveys with more than 15% missing data will be excluded from the final analysis to maintain data quality.
- Likert scale responses (1 = Strongly Disagree to 5 = Strongly Agree) will be assigned corresponding numerical values for statistical analysis.
- Demographic information (e.g., student year level) will be coded categorically.

Descriptive Statistics:

Purpose: To summarize the demographic profile of the student respondents and provide an overview of the current situation regarding student information access at ICCT Colleges.

Plan: Frequencies, percentages, means, and standard deviation will be calculated for:

- Demographic variables (year level, frequency of accessing school information).
- Current perceptions of information accessibility and clarity.
- Current levels of confusion regarding school document requirements and procedures.
- Frequency of student experiences with repetitive inquiries.
- Baseline measurements of student information uncertainty.

These descriptive statistics will establish a clear picture of the existing problems before system implementation.

Analysis of Information Fragmentation:

Problem: School document requirements procedures, and information are dispersed across multiple platforms, creating confusion among students at ICCT Colleges.

Quantitative Component:

The quantitative analysis of information fragmentation used frequency distribution and weighted mean analysis based on student responses to Section 1 of the UAT survey (Q1-Q5). Key indicators measured agreement levels on centralization, ease of finding document requirements

procedure, and reduction of fragmentation. The output was a numerical measure of the system's effectiveness in addressing information fragmentation.

Analysis of Repetitive Inquiries:

Problem: The lack of clear, accessible, and step-by-step requirement instructions compels students to ask the same questions multiple times, resulting in delayed transactions, repeated office visits, and student frustration.

Quantitative Component:

The quantitative analysis of repetitive inquiries will use frequency distribution and weighted mean analysis. Data will be sourced from student survey questions about their experiences with unclear instructions and repeated questions. Key indicators include the frequency with which students need to ask the same question, the number of office visits required to complete document requirements, and student perception of how unclear instructions affect their progress. The output will be a numerical measure of how often students experience repetitive inquiries and the resulting delays.

Analysis of Information Uncertainty:

Problem: Ambiguous guidelines and fragmented information sources create a cycle of uncertainty. Students, lacking clarity and trust in available information, are compelled to seek repeated confirmations.

Quantitative Component:

The quantitative analysis of information uncertainty will use weighted mean analysis. Data will be sourced from student survey questions about information clarity, trust, and frequency of seeking confirmations. Key indicators include self-reported uncertainty levels, confidence level in understanding school procedures, and frequency of checking multiple sources for the same information. The output will measure levels of student information uncertainty.

Purpose: To evaluate the effectiveness of the proposed eGuide ICCT system in addressing the three identified problems: information fragmentation, repetitive inquiries, and information uncertainty.

Quantitative Component:

Technique:

- Weighted mean analysis for usability and satisfaction ratings
- User Acceptance Testing (UAT) scoring

Source:

- Post-implementation evaluation survey
- User Acceptance Testing (UAT)

Evaluation Criteria:

- Usability: Ease of navigation, accessibility
- Information Fragmentation: Perception of having a centralized information source
- Repetitive Inquiries: Perceived reduction in repeated questions and office visits
- Information Uncertainty: Perceived reduction in student confusion

Output: Numerical rating of system effectiveness across multiple dimensions.

Statistical Treatment of Data

In order to determine the effectiveness of the eGuide ICCT system and to determine the answers to the three underlying problems raised by the study, fragmentation, irrelevant questions and uncertainty by developing solutions by using quantitative methods with formulas are implemented to data analysis that has been collected.

1. Weighted Mean

The weighted mean was employed in the calculation of the average score obtained from the student responses for every indicator under the 4 objectives of the study. This formula was computed in Section 1- 4 of the questionnaire survey used to calculate, degree of centralisation of information, how well do step- by step guidance help, trustworthiness and reliability of information, and overall usefulness and the users' tendency to use the system respectively.

The formula used is:

$$\bar{x} = \frac{\sum(f \cdot x)}{N}$$

Where:

- $\bar{x}$ = weighted mean
- f = frequency or number of responses for each rating
- x = numerical value assigned to each Likert scale response
- N = total number of respondents

2. Standard Deviation

Calculation The standard deviation was used to describe the variability and dispersion of responses about the average response. The formula will be used in addition with the weighted average mean derived for sections 1-4 to check whether the responses were similar for most of the students, or were there a considerable number of variations in the students.

The formula used is:

$$SD = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N - 1}}$$

Where:

- SD = standard deviation
- x_i = individual response value for each respondent
- $\bar{x}$ = mean of all responses for a particular item
- N = total number of respondents

3. Percentage

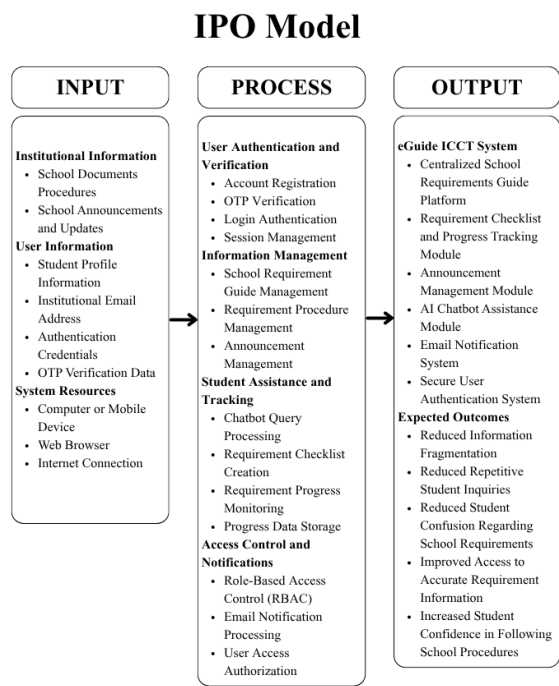
The percentage formula was used to compute the demographic profile of the respondents, the functional test pass rate from the User Acceptance Testing (UAT) results, and the overall UAT acceptance rate.

The formula used is:

$$\text{Percentage} = \left(\frac{\text{Frequency}}{\text{Total Number of Respondents}} \right) \times 100$$

Conceptual Framework / System Model

Figure 3.2: IPO Model



The conceptual framework of this study is based on the Input-Process-Output (IPO) Model, which illustrates how

the eGuide ICCT system transforms institutional information and user data into a centralized web-based guidance platform for students.

The Input phase consists of the information, data, and resources required for the operation of the system. These include school documents procedures, school announcements and updates, and guidance content. User-related inputs include student profile information, institutional email addresses, authentication credentials, and OTP verification data. Technical resources such as computers or mobile devices, web browsers, and internet connectivity are also necessary to access and utilize the platform effectively.

The Process phase describes how the system manages and transforms these inputs into functional services. The system performs account registration, login authentication, OTP verification, and session management to ensure secure user access. It also manages school requirement guides, requirement procedures, announcements, and knowledge-base content. In addition, the platform processes chatbot inquiries, creates and monitors requirement checklists, stores progress data, and implements Role-Based Access Control (RBAC), email notification processing, and user access authorization to support secure and efficient information delivery.

The Output phase is the eGuide ICCT system, a centralized school document requirements procedure guide platform that provides students with access to requirement procedures, progress tracking tools, announcements, chatbot assistance, email notifications, and secure user authentication features. The expected outcomes include reduced administrative overhead, reduced information fragmentation, reduced repetitive student inquiries, reduced student confusion regarding school document requirements procedure, improved access to accurate requirement information, and increased student confidence in following school procedures.

A feedback mechanism is incorporated into the framework to support continuous system improvement. User feedback, system evaluations, updated school procedures, and new institutional requirements may be used as inputs for future enhancements and content updates, ensuring that the platform remains relevant, accurate, and responsive to the needs of students and the institution.

Proposed System Overview

System Name: eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus

Purpose:

The system aims to collect and integrate the multiple academic information procedure into a single web application for students in ICCT Cainta Colleges. With accurate document information requirements procedures such as Summary Of Grade (SOG), the date of updating prospectus, and etc. System minimizing the inefficiencies of scattered information, repetitive in-person inquiries.

Key Features/Modules:

Centralized Documents Requirements Guide - Provides comprehensive, searchable information about academic document requirements procedure including enrollment procedures, document submissions, clearance processes, graduation requirements, and program-specific prerequisites organized by category and academic level.

User Registration and Verification - Enables new users to create an account by providing an email address and basic personal information. The system sends a one-time password (OTP) to the provided email for verification. The user must enter the OTP to complete registration and activate their account. This ensures that the email address is valid and accessible before granting access.

JWT-Based Session Management - Generates secure JSON Web Tokens upon successful authentication, allowing students to maintain authenticated sessions across the platform while browsing document requirements procedure and resources without repeatedly entering credentials.

Role-Based Access Control (RBAC) - Distinguishes between user types (Students, Administrators) and grants appropriate permissions based on assigned roles. Students access requirement guides and resources, while administrators manage content and user accounts.

Announcement Broadcasting System - Enables administrators to publish important notices, announcement, and procedural changes that are immediately visible to all authenticated users, ensuring students receive timely information about document requirements and school procedures.

Email Notification Services - Sends email alerts to students when administrators publish new announcements.

Administrative Content Management - Provides administrators with tools to create, update, and organize requirement guides, upload supporting documents, and maintain accurate, current information about academic and school procedures.

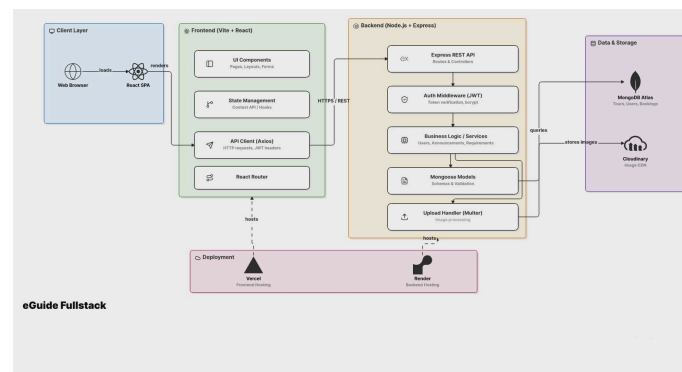
Target Users:

ICCT College Students - Primary users who require clear, accessible information about school document requirements

procedure, enrollment procedures, document submissions, clearance processes, and school policies to navigate their academic responsibilities efficiently

Administrative Staff - System administrators, registrar personnel, and student services staff responsible for maintaining accurate requirement information, managing user access, publishing updates, and responding to system-related concerns.

Figure 3.4: Architecture diagram



System Architecture

The system employs a modern full-stack web architecture consisting of:

Frontend Layer: React (Vite) framework providing a responsive single-page application with dedicated components for authentication (Login, OTP verification, Forgot Password), navigation (Navbar, AdminLayout), content display (Documents, Announcements, Homepage), and admin management (AdminDashboard, AdminDocuments, AdminAnnouncements).

Backend Layer: Node.js with Express framework implementing RESTful API services, including route handlers for authentication (authRoutes), user management (userRoutes), content management (announcementRoutes, requirementRoutes), saved progress (saveProgress), and an AI-assisted chatbot (chatbotRoutes).

Authentication Layer: JWT (JSON Web Token) implementation with Bearer token authorization, custom middleware (protect, adminOnly) for route-level access control, two-step email OTP verification on signup, and a 30-minute inactivity session timeout enforced on the frontend.

Business Logic Layer: Service modules handling email delivery (non-blocking queue), RAG-based chatbot retrieval with fuzzy and partial matching, Cloudinary image uploads, and in-memory response caching with automatic invalidation on data changes.

Data Persistence Layer: MongoDB Atlas cloud database with Mongoose ODM managing collections (User, Requirement, Announcement) through schema-based models with compound indexes for text search and query optimization.

Security Infrastructure: Password encryption via bcryptjs, role-based access control (student/admin), rate limiting on all API routes (express-rate-limit), CORS origin whitelisting, startup environment variable validation, input sanitization on chatbot queries, file type and size validation on uploads, and a global Express error handler ensuring consistent JSON error responses.

Summary

The eGuide ICCT system provides the ICCT community with a unified digital environment that addresses administrative delays and information fragmentation. The platform uses role-based access control and a modular architecture to maintain data security and accessibility, giving students 24/7 access to school document requirements procedure, announcements, and document guidance with AI-assisted chatbot. The deployment represents a strategic move toward a scalable, maintainable digital ecosystem built on current industry-standard technologies.

Requirements Analysis

Overview

You will get a focused guide for school document requirements procedure. The system reduces repeated inquiries and lowers student stress and worries. It protects institutional data and limits access to verified users. The design supports administrators who update content and send announcements

Functional Requirements

- Users can authenticate themselves by providing an email address and completing one-time password (OTP) verification.
- The system provides different access rights to students and administrators which are determined by their assigned roles.
- The system provides document requirements procedures which users can easily find through search functions.
- The document detail pages present both procedural details and information about the document.

- The administrative content management system enables administrators to create, edit, and delete document requirement procedures and announcement.
- The announcement module allows administrators to create announcements for students to receive.
- The system sends email notifications to users when administrators post new announcements and other system information.
- The web interface automatically adjusts itself to work with different web browsers and mobile devices.

Non-functional Requirements

- **Security** - The system implements security measures through JWT session management and encrypted password storage and role-based user access control and input data validation.
- **Reliability** - The system guarantees its accessibility during times when enrollment reaches its maximum capacity.
- **Usability** - The system is easy to navigate, accessible on both computers and mobile devices.
- **Performance** - The system architecture enables both user expansion and content growth to succeed.
- **Maintainability** - The staff admin interface allows staff members to perform content updates without needing any assistance from software developers.
- **Accessibility** - The system provides user-friendly design through its straightforward navigation system and visible search functionality and basic identification elements.
- **Privacy** - The system controls user access by permitting entry to verified users only while blocking access to users who do not have accounts.
- **Compliance** - The system provides support for current web browsers together with the most common screen display dimensions.
- **Limitations** - The system does not provide document storage capabilities and does not support grade and payment processing and lacks LMS functions and third-party authentication and mobile application support.

Table: 3.4: System Capabilities and Constraints

System Capabilities:	System Constraints:
The system authenticates users and authorizes their access using an email address and password.	The system is a document procedural guide only, it does not accept, store, or manage actual document submissions.
The system enables users to register by providing an email address and completing OTP verification.	The system is limited to providing a document procedural guide and posting announcement information. Functions outside these are not within the scope.
The system displays procedural guides which explain academic document requirements and enrollment procedures, etc. Administrators have the ability to create new document requirement information and update existing information.	All enrollment processes, fee payments, and financial record management are performed through separate Enrollment and Finance Systems and are outside the direct operational scope of this system.
The system broadcasts announcements to inform users about requirement changes and upcoming events and updates to procedures.	The official student portals serve as the main platforms through which students can complete their enrollment and registration and access their grades.
The system enables students to obtain requirement information at any time through any device that has internet access.	The system is currently a web-based application and does not have a native mobile application in its current version.
The system maintains user progress on the specific requirements or documents they are processing.	
The system organizes information according to two criteria: announcements, requirement and document category.	

Ethical, Legal, and Technical Considerations

The eGuide ICCT system is created to assist students. The system strives to ensure the accuracy and dependability of content, avoiding confusions, while minimizing student confusion and repetitive inquiries about document procedure. The system promotes equal access to information through an inclusive and user-friendly platform. All collected data will be used strictly for academic research purposes and will not be shared without user authorization. User access and system usage are subject to clearly specified terms and conditions to protect both the institution and its users.

This section discusses the technical aspects, requirements, and challenges addressed in developing and implementing the eGuide ICCT system for ICCT students. The system requires a mobile device or computer with sufficient processing memory, capability, and storage to ensure efficient operation. Users must have access to a reliable internet connection for stable performance. The system is developed using standard technologies and can be accessed through common web browsers such as Google Chrome and Microsoft Edge. It is built to support different devices and operating systems. System performance may vary depending on device specifications and browser versions. Primary

security measures, including user authentication and proper data management, are implemented to protect information and maintain data confidentiality. Despite these provisions, applications may face limitations such as internet dependency, server uptime, legacy software, and compatibility issues with older devices.

CHAPTER 4 RESULTS AND DISCUSSION

The primary purpose of this chapter is to present the findings from the implementation of the eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus. Three significant problems at ICCT Colleges Main Campus were the focus of this study: information fragmentation, repetitive inquiries, and information uncertainty. The results presented in this chapter are based on quantitative data collected from 160 student respondents who took part in Likert-scale surveys and post-implementation User Acceptance Testing (UAT) after the system was deployed. This chapter is organized according to the four specific objectives of the study: centralization of fragmented information, provision of step-by-step guidance, delivery of precise and consistent information, and overall evaluation of usability and effectiveness. Each section restates the objective, presents the corresponding results in tabular form, and provides a detailed discussion of the findings, their implications, and how they address the original problems identified at ICCT Colleges.

A. Respondent / Dataset Profile

Table 4.1: Respondent Demographics

Year Level	Frequency	Percentage
1st year	24	15%
2nd year	38	23.7%
3rd year	85	53.1%
4th year	13	8.1%
Total:	160	100%

Out of the 160 students who participated, third year was the largest group with 86 respondents (53.42%), This was followed by students in their second year with 38 respondents (23.60%). First-year students accounted for 24 respondents (14.91%), while fourth-year students represented the smallest group with 13 respondents (8.07%).

B. Results for Objective 1: Centralize Fragmented Information

To centralize fragmented requirement information from multiple sources (Facebook, bulletin boards, word-of-mouth) into a single, organized digital platform.

Table 4.2: Weighted mean results (Section 1: Q1–Q5)

Question/ item Q1-Q5	Mean	SD	Interpretation
1. eGuide ICCT centralizes all school requirement information in one platform.	4.48	0.61	Strongly Agree
2. I can easily find all document requirement information without checking multiple sources (Facebook, bulletin boards, etc.).	4.19	0.73	Agree
3. The system eliminates the problem of requirement posts getting buried or lost.	4.41	0.61	Strongly Agree
4. All document requirement information in eGuide ICCT is consistent across different pages and sections.	4.24	0.63	Strongly Agree
5. Compared to previous methods, eGuide ICCT has significantly reduced information fragmentation.	4.43	0.61	Strongly Agree
Weighted Average Mean & SD	4.35	0.64	Strongly Agree

Discussion:

Table 4.2 Weighted Means for the construct of Information Fragmentation (Q1- Q5, Part 1) for the students who responded n=160. The statement “eGuide ICCT centralizes all school requirement information in one platform.” has the highest weighted mean 4.48 (SD=0.61)

with the description “Strongly Agree,” and the statement “I can easily find all document requirement information without checking multiple sources.” got lowest mean 4.19 (SD =0.73) which means “Agree.” Weighted Mean=4.35 (SD =0.64) implies that respondents strongly agree with the centralizing fragmented information, especially that “eGuide ICCT centralizes all school requirement information in one platform”. Since SD=0.64, respondents' responses in average range are close to one another, meaning results are trustworthy and meaningful. This finding seems in line with the study of Pinho, Franco and Mendes [5] which states the advantages of having a central portal which gives students access to information and other resource centres.

C. Results for Objective 2: Step-by-Step Guidance for Repetitive Inquiries

To provide clear, step-by-step guidance that enables students to complete document requirements without repeated office visits.

Table 4.3: Weighted mean results (Section 2: Q6–Q10)

Question/ item Q6-Q10	Mean	SD	Interpretation
6. eGuide ICCT provides clear, step-by-step instructions that I can follow without staff assistance.	4.42	0.63	Strongly Agree
7. Using eGuide ICCT would reduce the number of times I need to visit an office for requirement questions.	4.18	0.59	Agree
8. The system helps me submit complete document requirements on my first attempt.	4.2	0.60	Agree
9. eGuide ICCT has reduced the need to ask the same question multiple times.	4.31	0.63	Strongly Agree
10. The system reduces the number of times I need to ask the same question,	4.29	0.60	Strongly Agree

allowing me to complete document requirements more efficiently.			
Weighted Average Mean & SD	4.28	0.61	Strongly Agree

Discussion:

Table 4.3 reports weighted mean findings for Repetitive Inquiries (Section 2, Q6-Q10) among 160 student-respondents. The item "eGuide ICCT provides clear, step-by-step instructions that I can follow without staff assistance" had the highest weighted mean value of 4.42 (SD = 0.63), scaled as "Strongly Agree", while the item "Using eGuide ICCT would reduce the number of times I need to visit an office for requirement questions" received the lowest mean score of 4.18 (SD = 0.59), scaled as "Agree". With an overall mean of 4.28 (SD = 0.61) that falls within the "Strongly Agree" scale, respondents have strongly supported eGuide ICCT in its function of giving clear and step-by-step instructions which will minimize Repetitive Inquiries and office visits. The low standard deviation of 0.61 confirms consistency among respondents in responding to this dimension which enhances validity of findings. This is consistent with work of Muso [7] that indicated that web-based academic information systems can help by having clear instructions that guide the user online to improve both staff and students' efficiency.

D. Results for Objective 3: Accurate, Reliable, Consistent Information

To ensure accurate, reliable, and consistent information that reduces student confusion and uncertainty.

Table 4.4: Weighted Mean Results (Section 3, Q11-Q20)

Question/item Q11-Q20	Mean	SD	Interpretation
11. I trust the requirement information in eGuide ICCT is accurate and official	4.45	0.60	Strongly Agree
12. eGuide ICCT has reduced my confusion about how to interpret requirement instructions.	4.31	0.61	Strongly Agree

13. I trust eGuide ICCT more than information found on Facebook or from other students.	4.76	0.49	Strongly Agree
14. eGuide ICCT has reduced inconsistent submissions caused by students relying on different sources.	4.19	0.64	Agree
15. Overall, eGuide ICCT has significantly reduced information uncertainty and confusion at ICCT	4.53	0.63	Strongly Agree
Weighted Average Mean & SD	4.45	0.59	Strongly Agree

Discussion:

The weighted mean analysis for Information Uncertainty (Section 3, Q11-Q15) is detailed in Table 4.4, which represents the analysis based on responses of 160 student-respondents. The item "I trust eGuide ICCT more than information found on Facebook or from other students" obtained the highest weighted mean of 4.76 (SD = 0.49) interpreted as Strongly Agree, and the item "eGuide ICCT has reduced inconsistent submissions caused by students relying on different sources" recorded the lowest mean of 4.19 (SD = 0.64) interpreted as Agree. The overall weighted mean of 4.45 (SD = 0.59) interpreted as Strongly Agree indicates that the eGuide ICCT had strongly helped in delivering accurate and consistent and reliable information that minimises the uncertainty and confusion of the students. The SD of 0.59 of overall respondents shows uniformity of replies in all the students, hence proving the authenticity of these results. This is in support with the finding of Laugi [19], which had provided the benefit of having centralised systems increased the transparency and assured students in obtaining accurate and consistent information about the procedures of academic operations.

E. Results for Objective 4: Usability and Effectiveness Evaluation

To evaluate the usability and effectiveness of the system at ICCT Main Campus.

Table 4.5: Willingness to Use Results (Section 4)

Question/item Q17-Q19	Weighted Mean	SD	Interpretation
16. 16. eGuide ICCT would be a valuable addition to ICCT's current information dissemination process.	4.47	0.63	Strongly Agree
17. Compared to current methods (Facebook, bulletin boards, word-of-mouth), this system is more effective in ensuring students receive accurate requirement information.	4.4	0.62	Strongly Agree
18. The system makes it easier to disseminate information and requirement updates to students.	4.46	0.62	Strongly Agree
19. I would actively use and recommend eGuide ICCT to other students or administrators at ICCT.	4.48	0.58	Strongly Agree
Weighted Average Mean & SD	4.45	0.61	Strongly Agree

Discussion:

In Table 4.5, shows the weighted mean on willingness to use the System (Section 4 Q16 – Q19) based on responses of 160 student-respondents. The question; "I will surely make the use of eGuide ICCT and surely promote it to my co-students and administration in ICCT." had the highest weighted mean of 4.48 with SD of 0.58 which means "Strongly Agree," while the question; "It is effective with respect to delivering adequate requirement's' information compared to present methodology like (Facebook, Bulletin Board, Word-of- Mouth)" the lowest weighted mean of 4.4 with SD of 0.62, also interpreted as "Strongly Agree." The total mean of 4.45 with SD of 0.61 interpreted as "Strongly Agree" is a testament to the readiness of ICCT students in the implementation of eGuide ICCT. The standard deviation of 0.61 suggests that most of the responses are close to each other and it represents uniformity of the response of every respondent, consistent with Caratiquit [12] who suggests web

based school information's help delivery of announcements & learning material. Javelosa et al [16] likewise found out that information web systems improves access, ease, efficiency and automation of management of operations depending on roles and responsibilities.

Table 4.6: Functional Testing Results Summary

Question/Item F1 - F16	Pass (n)	Fail (n)	Pass (%)
F1. I was able to log in from the system	160	0	100%
F2. I was able to access the system without errors.	159	1	99.38%
F3. I was able to view school document requirements and procedures.	160	0	100%
F4. I was able to view step-by-step guides.	159	1	99.38%
F5. I was able to search for specific document requirements procedures.	159	1	99.38%
F6. I was able to see announcements and updates.	160	0	100%
F7. I received email notifications when admin posted announcements	157	3	98.13%
F8. I was able to download/save document requirements procedure as PDF.	159	1	99.38%
F9. The system worked properly on my device.	160	0	100%
F10. I was able to search for specific announcements and updates.	160	0	100%
F11. The system showed an appropriate message when my search had no	157	3	98.13%

results.			
F12. I was able to create a new account (sign up) successfully.	159	1	99.38%
F13. I received an OTP (one-time password) when I signed up.	160	0	100%
F14. I was able to reset my password using the "Forgot Password" feature.	160	0	100%
F15. I received an OTP when I requested to reset my password.	160	0	100%
F16. I was able to log in using my new password after resetting it.	160	0	100%
Average:			99.57%

Discussion:

Table 4.6 shows that all functional tests (F1 to F16) achieved an average pass rate of 99.57%. Key functions including login, viewing document requirements procedure, search, notifications, PDF download, account creation, OTP receipt, and password reset all passed successfully. The consistently high pass rates demonstrate that the eGuide ICCT system is functioning reliably as intended.

G. Exploratory Feedback from Administrative Staff

Table 4.7: Summary of Admin Feedback

Question	Feedback
How easy or difficult was it to navigate the admin dashboard?	"The system is easy to navigate."
How straightforward was the process of posting announcements and updates?	"The process is straightforward and user-friendly."
How easy was it to add, edit, or delete requirement guides?	"It was easy to add, edit, and delete requirement guides."
Overall, how satisfied are you with the admin dashboard?	"Yes, I am satisfied."
Are there any security concerns you would like to raise?	"The administrative staff suggested strengthening security measures. The system already implements JWT-based authentication, role-based access control (RBAC),

	and password encryption, but their feedback is noted for continuous improvement."
What is your overall impression of the eGuide ICCT system?	"The system is good. Informative, easy to navigate, and the design is nice."
What features or improvements would you like to see in future versions of the system?	"Use student number instead of email for login. Admin should have the ability to edit or recover student accounts to prevent duplicates and reduce storage."
Do you have any suggestions for improving the login process?	"Suggestion to use student number instead of email for login."

Discussion:

The 3 administrative staff responded positively, noting that the system was easy to navigate, informative, and well-designed. They expressed overall satisfaction. Key suggestions included using student numbers for login and adding admin account recovery to prevent duplicate accounts and reduce storage. Regarding security, the system already implements JWT, RBAC, and password encryption, but their feedback is noted for continuous improvement.

H. Integrative Discussion

How findings collectively address all three problems:

Problem 1: Information Fragmentation – Table 4.2 (Mean = 4.35, SD = 0.64) confirms that eGuide ICCT centralizes requirement information previously scattered across Facebook, bulletin boards, and word-of-mouth. Q1 (centralizes all school requirement information in one platform, Mean = 4.48) directly indicates that students now have a single, unified source for their requirements, solving the core issue of information fragmentation.

Problem 2: Repetitive Inquiries – Table 4.3 (Mean = 4.28, SD = 0.61) demonstrates that step-by-step guidance reduces repeated questions and office visits. Q9 (reduced the need to ask the same question multiple times, Mean = 4.31) and Q7 (reduces office visits, Mean = 4.18) provide direct evidence that the system shifts students toward self-service access, though Q7's slightly lower rating suggests that some students still need to visit offices occasionally.

Problem 3: Information Uncertainty – Table 4.4 (Mean = 4.45, SD = 0.59) demonstrates that accurate, reliable information reduces student uncertainty. Notably, Q13 (trust eGuide more than Facebook, Mean = 4.76, SD = 0.49) is the highest rating across all objectives, reflecting a complete reversal of pre-system reliance on unofficial, conflicting sources.

Patterns across objectives:

Trust over informal sources – The exceptional 4.76 mean for Q13 indicates that students now consider eGuide ICCT more authoritative than Facebook or peer advice, a critical cultural shift at ICCT.

Consistency needs minor improvement – Q4 (consistency across pages, Mean = 4.24) and Q14 (reducing inconsistent submissions, Mean = 4.19) scored slightly lower, suggesting room for improvement in cross-page content standardization and continued system adoption.

Strong willingness to recommend – Section 4 (Mean = 4.45, SD = 0.61) and Q19 (would actively use and recommend, Mean = 4.48) indicate high potential for organic adoption.

Functional reliability builds trust – The 99.57% functional test pass rate provides a technical foundation for high subjective ratings.

Admin validation – Staff confirmed ease of navigation and overall satisfaction, offering practical suggestions (student number login, account recovery) while noting that existing security measures (JWT, RBAC) meet baseline standards.

Overall contribution:

eGuide ICCT is a transformative solution for requirement information management at ICCT Main Campus. Based on data from 160 students, key contributions include:

- Reduced student repetitive inquiries – Fewer repeated office visits (Section 2 Mean = 4.28)
- Improved student experience – Centralized, trustworthy information (Section 1 Mean = 4.35; Section 3 Mean = 4.45)
- Cultural shift – Students now trust eGuide over Facebook (Mean = 4.76)
- Ready for deployment – 99.57% functional pass rate

I. Limitations of the Findings

The quantitative and qualitative results presented in this study demonstrate strong user approval and system viability for the eGuide ICCT system. However, to ensure an objective academic evaluation, it is essential to outline the inherent boundaries, constraints, and limitations encountered during the research process. Recognizing these boundaries provides a realistic context for the findings and establishes a baseline for future research.

A. Sample Constraints

The primary limitation concerns the study's participant demographic and institutional boundaries. The empirical data collected during this study was restricted to a sample size of 160 student respondents explicitly enrolled at the ICCT Colleges Main Campus. Because the sample was drawn exclusively from a single campus environment, the demographic profile, student behaviors, and localized administrative workflows may not fully represent the diverse student populations or operational challenges across alternative branch campuses or distinct higher education institutions. The high consensus observed in the data reflects the specific academic climate of the Main Campus rather than a universal baseline for web-based academic guiding platforms. Also the total population of students in ICCT Cainta is unknown since the school does not fully know the entire population in the main campus.

B. Environmental Factors

External, systematic, and real-world environmental elements during the deployment phase may have influenced user responses. As a centralized web-based application accessible through browsers on computers and mobile devices, the operational performance and perceived efficiency of eGuide ICCT heavily rely on steady internet connectivity and local network stability. Fluctuations in data access or campus Wi-Fi availability during the deployment and testing phases could alter user perceptions of system speed, responsiveness, and notification reliability, independent of the actual software architecture. The evaluation was conducted shortly after initial system deployment; the positive feedback regarding notifications may have been partially influenced by the "novelty effect," where users rate a new platform highly simply because it replaces tedious, traditional processes like physical bulletin boards or fragmented social media posts.

C. Measurement Limitations

The methodologies and tools used to capture and analyze data present their own set of analytical boundaries. The evaluation relies heavily on closed-ended, quantitative instruments, specifically Likert-scale surveys and post-implementation User Acceptance Testing (UAT). While descriptive statistics like weighted means and standard deviations effectively prove a strong consensus among respondents, self-reported data can be subject to response bias, where students might overstate their positive feedback due to the perceived desirability of a system.

D. Scope Boundaries

The technical and functional boundaries deliberately set during the development of eGuide ICCT restrict the direct application of the findings. Structurally, the notification type is limited to sending emails only when administrators post announcements and does not provide real-time messaging or live chat alerts. The system is designed strictly as an informational and procedural guide about school document requirements procedure and announcements; it is not a

transactional application and does not integrate with official registrar databases or policy-making modules. The system was built and tested strictly as a web-based application; the findings cannot be generalized to deduce how users would interact with a native mobile application environment or a purely offline infrastructure.

E. Document Coverage Limitations

Due to the time constraints of the capstone project and limited access to complete institutional records, the researchers were only able to gather requirement documents and procedures that were readily available from existing sources. These sources included Facebook posts from official ICCT pages and groups, physical bulletin boards around the campus, and shared files from department offices. As a result, the system does not contain all existing ICCT requirement documents and procedures. The document library represents the collection of what the research team was able to obtain within the project timeline. However, the system architecture is designed to support the addition of more documents and procedures in future implementations.

CHAPTER 5 SUMMARY, CONCLUSION, AND RECOMMENDATION

A. Summary of Findings

This research developed and assessed the eGuide ICCT: A Centralized Web-Based Guide System for School Document Procedures at ICCT Cainta Main Campus. The system aimed to resolve three interrelated challenges at ICCT Colleges: scattered information, repeated student inquiries, and unclear or unreliable guidance. Following system deployment, data were gathered from 160 student participants using Likert-scale surveys and User Acceptance Testing (UAT).

Objective 1: Centralize Fragmented Information

The system garnered a weighted mean of 4.35 (SD = 0.64) for its role in unifying scattered requirement information, which translates to a "Strongly Agree" rating. Among the indicators, Q1 ("eGuide ICCT centralizes all school requirement information in one platform") scored the highest, with a mean of 4.48 (SD = 0.61) a clear indication that students firmly believed the platform effectively consolidated all requirements into a single, cohesive hub. In contrast, Q2 ("I can easily find all document requirement information without checking multiple sources") recorded the lowest average at 4.19 (SD = 0.73), placing it in the "Agree" range. Although still a favorable rating, this comparatively lower score implies that a minority of students occasionally turned to other channels, most likely due to the limited document library available during the pilot phase. The

remaining items also performed strongly: Q3 ("The system eliminates the problem of requirement posts getting buried or lost") posted a mean of 4.41 (SD = 0.61), Q4 ("All document requirement information is consistent across pages") averaged 4.24 (SD = 0.63), and Q5 ("Compared to previous methods, eGuide ICCT has significantly reduced information fragmentation") reached 4.43 (SD = 0.61). Importantly, all indicators comfortably surpassed the predetermined success benchmark of 3.41, confirming that the overall Achievement Status is **ACHIEVED**.

Objective 2: Step-by-Step Guidance for Repetitive Inquiries

The evaluation yielded an overall weighted mean of 4.28 (SD = 0.61), falling under "Strongly Agree," which indicates that well-structured, sequential guidance allowed students to fulfill their document requirements without needing repeat visits to campus offices. The top-performing indicator was Q6 ("eGuide ICCT provides clear, step-by-step instructions that I can follow without staff assistance"), with a mean of 4.42 (SD = 0.63). Q9 ("eGuide ICCT has reduced the need to ask the same question multiple times") followed closely at 4.31 (SD = 0.63), while Q10 ("The system reduces the number of times I need to ask the same question, allowing me to complete document requirements more efficiently") posted 4.29 (SD = 0.60). Meanwhile, Q8 ("The system helps me submit complete document requirements on my first attempt") averaged 4.2 (SD = 0.60), interpreted as "Agree," and Q7 ("Using eGuide ICCT would reduce the number of times I need to visit an office for requirement questions") recorded 4.18 (SD = 0.59), also in the "Agree" range. These relatively modest scores for Q7 and Q8 suggest that, although students generally acknowledged the system's benefits in cutting down office visits and improving first-attempt submissions, there remains room for enhancement—especially given that the platform currently does not interface with the registrar's database to offer real-time, student-specific data. Nonetheless, all items exceeded the success threshold, and the overall Achievement Status is **ACHIEVED**.

Objective 3: Accurate, Reliable, Consistent Information

yielded a weighted mean of 4.45 (SD = 0.59), also interpreted as "Strongly Agree," indicating that dependable and trustworthy content effectively reduced student uncertainty. Notably, the strongest rating across all objectives came from Q13 ("I trust eGuide ICCT more than information found on Facebook or from other students"), which achieved a remarkable mean of 4.76 (SD = 0.49)—a clear signal of a significant shift away from informal, peer-based sources toward the system as a credible reference. Q15 ("Overall, eGuide ICCT has significantly reduced information uncertainty and confusion at ICCT") followed with a mean of 4.53 (SD = 0.63). Meanwhile, Q11 ("I trust the requirement information in eGuide ICCT is accurate and official") recorded 4.45 (SD = 0.60), and Q12 ("eGuide ICCT has

reduced my confusion about how to interpret requirement instructions") posted 4.31 (SD = 0.61). The lowest of the five was Q14 ("eGuide ICCT has reduced inconsistent submissions caused by students relying on different sources"), with a mean of 4.19 (SD = 0.64), falling under "Agree." This relatively modest score suggests that, although the system has made headway in minimizing submission discrepancies, there is still scope for refinement—particularly as the document library continues to grow and as more students come to rely on eGuide ICCT as their primary information hub. All indicators surpassed the 3.41 benchmark, so the Achievement Status remains **ACHIEVED**.

Objective 4: Usability and Effectiveness Evaluation

The fourth section, which assessed willingness to use the system, registered a weighted mean of 4.45 (SD = 0.61), corresponding to a "Strongly Agree" interpretation. The top performer was Q19 ("I would actively use and recommend eGuide ICCT to other students or administrators at ICCT"), with a mean of 4.48 (SD = 0.58), pointing to a strong likelihood of organic uptake and peer-driven advocacy among the student body. Other items also scored highly: Q16 ("eGuide ICCT would be a valuable addition to ICCT's current information dissemination process") posted 4.47 (SD = 0.63), Q18 ("The system makes it easier to disseminate information and requirement updates to students") averaged 4.46 (SD = 0.62), and Q17 ("Compared to current methods such as Facebook, bulletin boards, word-of-mouth, this system is more effective") recorded 4.4 (SD = 0.62). On the technical side, functional testing yielded an impressive average pass rate of 99.57% across 16 test scenarios, which covered core functionalities including login, document procedure viewing, search, notifications, PDF downloads, account creation, OTP verification, and password reset. Qualitative input from administrative staff described the platform as "easy to navigate," "informative," and "well-designed," though they also offered suggestions for future refinements—such as enabling student number-based login and bolstering account recovery options. With all criteria met, the Achievement Status is **ACHIEVED**.

B. Conclusions

Conclusion for Objective 1: Centralization of Fragmented Information

The eGuide ICCT system has brought about a fundamental transformation in the way students at ICCT Cainta Main Campus obtain their requirement information. Prior to its deployment, students relied on a disjointed mix of Facebook updates, physical notice boards, and word-of-mouth exchanges, which often bred confusion and frequently resulted in critical details being missed or buried. The Section 1 weighted mean of 4.35 (SD = 0.64) provides quantifiable evidence that the system effectively eliminated the necessity of navigating several different platforms. This result aligns

with the conclusions of Nkomo, Daniel, and Butson [1], who demonstrated that centralized web portals significantly enhance students' access to academic information. Similarly, Pinho, Franco, and Mendes [5] noted that such platforms improve information management and ease communication flows in higher education settings. The high rating for Q1 (Mean = 4.48) directly corroborates that students currently enjoy a single, consolidated repository for their needs, effectively tackling the central problem of information fragmentation that this study sought to address. Although Q2 (Mean = 4.19) garnered a comparatively modest score, this is largely ascribed to the restricted document repository during the initial trial phase, which was confined to materials the research team could collect within the project's timeframe a constraint that can be remedied by persistently broadening the content base.

Conclusion for Objective 2: Reduction of Repetitive Inquiries

The step-by-step guidance built into eGuide ICCT effectively minimized the recurrence of student inquiries and cut down on unnecessary trips to campus offices. Before the system was in place, the absence of standardized, easily retrievable instructions meant that students often had to ask the same questions repeatedly, visit offices multiple times, and submit incomplete paperwork. The Section 2 weighted mean of 4.28 (SD = 0.61) confirms that students broadly agreed the system addressed these issues. This finding echoes the work of Muso [7], who found that web-based academic information systems with clear, online procedures improve efficiency for both staff and students. Similarly, Sukma et al. [21] demonstrated that online guidance platforms offer more convenient and structured access, thereby reducing scheduling conflicts and incomplete submissions. The strong scores for Q6 (clear instructions without staff assistance, Mean = 4.42) and Q9 (fewer repeated questions, Mean = 4.31) reinforce that eGuide ICCT successfully directed students toward self-service access. The somewhat lower ratings for Q7 (fewer office visits, Mean = 4.18) and Q8 (complete on first attempt, Mean = 4.2) point to the need for further refinement specifically, integration with the registrar's database to deliver real-time, student-specific data remains absent, which may still require some students to consult offices occasionally. Even so, the overall outcomes reflect tangible progress in lessening administrative burdens while simultaneously improving the student experience.

Conclusion for Objective 3: Accurate, Reliable, and Consistent Information

By offering a credible and official repository for requirement-related information, eGuide ICCT effectively lessened students' uncertainty about their academic tasks. In the previous setup, students had no reliable way to confirm whether details shared via Facebook or by peers were accurate or sanctioned, which often led to inconsistent submissions and

a heavy reliance on staff verification. The Section 3 weighted mean of 4.45 (SD = 0.59)—the highest among all objectives—offers strong evidence that the system successfully addressed this challenge. Of particular note, Q13 ("I trust eGuide ICCT more than information found on Facebook or from other students," Mean = 4.76) stood out as the single highest-rated item across the entire survey, signaling a marked transformation in student information-seeking behavior. This aligns with the observations of Zhiyong et al. [6], who contended that digital platforms boost user satisfaction and institutional effectiveness when content is well-structured and processes are streamlined. The results also echo the work of Laugi [19], who found that centralized systems enhance transparency and ensure that students receive accurate, uniform information regarding academic procedures. The somewhat lower score for Q14 (reducing inconsistent submissions, Mean = 4.19) implies that, although the system has made headway, further expansion of the document repository and wider student uptake will continue to diminish dependence on divergent sources. Ultimately, the exceptionally high trust rating affirms that eGuide ICCT has established itself as the go-to, authoritative source for requirement information at ICCT Cainta Main Campus.

Conclusion for Objective 4: Usability and Effectiveness

The eGuide ICCT system exhibited strong usability, dependable functionality, and broad user acceptance at ICCT Main Campus. This is substantiated by a willingness-to-use weighted mean of 4.45 (SD = 0.61) and a functional testing pass rate of 99.57%. The high advocacy score for Q19 (mean = 4.48) further underscores that students recognize the platform's value and are inclined to promote its use among peers. These results are in line with the work of K. D. Caratiquit [12], who observed that web-based school information systems accelerate the dissemination of announcements and learning materials, as well as with Javelosa et al. [16], who showed that such systems enhance accessibility, efficiency, and automated task management tailored to different user roles. Administrative staff provided qualitative validation, praising the system for being intuitive, informative, and visually well-crafted, while also offering constructive suggestions such as implementing student number-based login and strengthening account recovery mechanisms, which can be incorporated in future updates. The functional test outcomes confirm that the system reliably executes its core operations, establishing a sound technical basis for sustained institutional integration over the long term.

Overall Integrative Conclusion

The design and assessment of the eGuide ICCT system offer a unified and impactful solution to the interlinked challenges of fragmented information, redundant queries, and data uncertainty at ICCT Cainta Main Campus. Although each objective was independently met, the broader value of this

study lies in how these successes combine synergistically to produce a truly transformative outcome.

The synthesis begins with Objective 1 (Centralization), which established a consolidated digital repository (Weighted Mean = 4.35, SD = 0.64), effectively breaking down the previously scattered ecosystem of Facebook posts, bulletin boards, and informal conversations. This centralized hub laid the essential groundwork for Objective 2 (Step-by-Step Guidance), which translated the aggregated content into coherent, actionable sequences (Weighted Mean = 4.28, SD = 0.61). By offering a single, definitive source of truth paired with clear instructions, the system enabled students to handle document requirements without unnecessary office visits, thereby interrupting the cycle of recurring questions.

The platform's credibility was further reinforced by Objective 3 (Accuracy & Reliability), which drew the strongest user ratings across all objectives (Weighted Mean = 4.45, SD = 0.59). The unparalleled trust students placed in eGuide ICCT over informal sources (Q13, Mean = 4.76) did not occur in isolation; it arose directly from the system's consistent, transparent, and dependable information. This trust effectively resolved the pre-existing uncertainty, as students no longer felt compelled to seek expensive and time-consuming validation from multiple channels.

Finally, Objective 4 (Usability & Effectiveness) confirmed that this integrated solution is not only conceptually robust but also practically viable and ready for implementation. The 99.57% functional test pass rate and acceptance by all 160 UAT respondents demonstrate that the technical infrastructure reliably supports the combined benefits of centralization, clarity, and accuracy. The strong inclination to use and advocate for the system (Section 4, Mean = 4.45, SD = 0.61) indicates that students clearly recognize the comprehensive value of this unified platform.

The integrative discussion in Chapter 4 further reinforces these conclusions. The findings collectively resolve all three core issues: information fragmentation (Section 1 Mean = 4.35), repetitive inquiries (Section 2 Mean = 4.28), and information uncertainty (Section 3 Mean = 4.45). The exceptional Q13 rating (Mean = 4.76) signals a complete departure from the earlier reliance on unofficial, conflicting sources—a significant cultural shift at ICCT. While Q4 (content consistency across pages, Mean = 4.24) and Q14 (reduction in inconsistent submissions, Mean = 4.19) scored somewhat lower, indicating room for refinement in cross-page standardization, the overall results nevertheless confirm that the system has successfully tackled the principal challenges identified in this study.

In summary, the eGuide ICCT system effectively addresses the identified problems not as separate issues, but as a single systemic challenge of information disorder. Through content centralization, procedural clarity, reliability assurance, and

usability validation, this research has delivered a functional, trusted, and effective digital framework. It demonstrably improves the student experience, reduces administrative burdens, and offers ICCT Cainta Main Campus a scalable model for modernizing academic information dissemination. Since all four specific objectives have been fully achieved, the overall general objective of the study has likewise been accomplished.

C. Recommendations

1. Recommendations for Practice

A protocol for continuous content validation is being implemented. Items pertaining to cross-page consistency (Q4, Mean = 4.41) and minimizing inconsistent contributions (Q14, Mean = 4.37) earned significantly lower, but still good, ratings, despite the research reporting high user trust (Mean = 4.60 for accuracy) and substantial agreement on decreased fragmentation (Mean = 4.47). It is advised that the administrative staff set up a regular content audit procedure (every two weeks, for example) to solve this. This protocol should include a checklist to ensure that all related requirement pages (e.g., enrollment procedure for different departments) display consistent information, version control dates are visible, and all supporting documents are the current school-issued copies. This practice will directly strengthen the "consistent information" objective.

Adoption of Student Number-Based Authentication and Account Recovery Workflow. A practical issue was brought to light by qualitative input from administrative staff: students who presently forget their password or registered email frequently establish new accounts rather than retrieving their old ones, which might cause database bloat and confusion. Instead of using an email address as the primary identification, it is advised that the login procedure be changed to accept the official ICCT student number. In order to ensure data integrity and minimize repeated user entries, an account recovery mechanism designed only for administrators should be put in place. This will enable them to confirm a student's identification and restore credentials for a single, canonical account.

2. Recommendations for Future Researchers

The current study was limited to ICCT Cainta Main Campus. Future research should expand the scope to include other ICCT campuses and a wider range of student populations. This would determine whether the findings generalize across different campus environments, student demographics, and administrative workflows.

This study was conducted exclusively at the ICCT Cainta Main Campus, with a sample size of 160 students drawn from this single location. While the system architecture was designed to support potential expansion, the

generalizability of the findings to other ICCT campuses remains untested. Future researchers are encouraged to implement and evaluate the eGuide ICCT system at other ICCT branch campuses, such as Antipolo, Taytay, or San Mateo.

The current study gathered qualitative feedback from a small sample of administrative staff through open-ended discussions. Future research should conduct a formal administrator survey with a larger sample to provide quantitative data on admin perceptions, dashboard usability, and content management efficiency. This would complement the student-focused findings and provide a more comprehensive evaluation of the system.

While the current eGuide ICCT system includes a RAG-based chatbot and fuzzy matching for information retrieval, there are opportunities to integrate more advanced technologies to further enhance system intelligence and user experience. Future researchers are recommended to explore:

Enhanced Chatbot Capabilities – Improving the current chatbot by adding session-based memory so it can handle follow-up questions, and expanding its knowledge base to cover more topics based on frequently asked questions from students. This would make the chatbot more helpful and reduce the need for students to ask repetitive questions.

3. Recommendations for ICCT Colleges / Institution

The current study measured immediate user perceptions after system deployment. Future research should conduct a longitudinal study to determine whether positive ratings persist over multiple academic semesters and whether the system continues to reduce repetitive inquiries over time. This would provide valuable insights into the system's long-term sustainability and user acceptance.

While the eGuide ICCT system proved effective in helping students, the current document library only contains the information the research team was able to gather within the project timeline. To make the system more complete, ICCT Colleges should systematically collect and upload all requirement documents and procedures from all departments, create a clear process for writing new guides, use standard templates to ensure consistency, and assign specific staff or departments to keep the information accurate and up to date. These steps will ensure that eGuide ICCT remains a reliable and complete source of guidance for all students.

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Appendix

Appendix A: Letter of Approval

Dear ICCT Colleges,

Good day,

We are students currently undertaking our capstone project, and would like to request the proper procedure for obtaining all school requirements or documents (e.g., Prospectus, SOG, Good Moral). Specifically, we seek clear guidance on the step-by-step process, required documents, and any policies or protocols that students must follow.

Based on our survey results, many students encounter difficulties accessing important information. In particular, 78.3% (55% agree, 23.3% strongly agree) reported that information from platforms like Facebook is often disorganized or hard to find. Additionally, 86.8% (47.5% agree, 39.3% strongly agree) believe that having all school requirements in one system would save time. Moreover, 85.3% (44.3% agree, 41% strongly agree) prefer using a web-based system rather than asking multiple individuals for information. These findings emphasize the need for a more organized and accessible system.

This request will greatly contribute to the accuracy and relevance of our capstone project, which aims to improve students' access to academic services and school-related information.

We sincerely hope for your assistance and guidance. Thank you for your time and consideration.

Respectfully,

Alday, John Denrick M.
Canuel, Mark Justin R.
Eusebio, Rain Justine R.
Familara, Carl Justin B.
Gaoat, Athena Jem D.
Huerto, Benlester N.
Laguetao, Maria Angela B.
Parale, Carl Kristian A.
Singian, Em-Jay P.
Verdan, Geryk Drew P.


Jerico Villog

Appendix B: Informed Consent Form

Dear Respondent,

Good day!

We are a 3rd year Bachelor of Science in Information Technology (BSIT) students from ICCT Colleges currently conducting a research study entitled "eGuide ICCT: A Centralized Web-Based School Requirements Guide System with Automated Notifications."

In line with this, we would like to respectfully request your permission to participate in an interview for our research study. The interview aims to gather valuable insights and experiences related to school requirements, procedures, and information dissemination within the institution.

Please be assured that your participation is completely voluntary. All information and responses that you will provide will be treated with strict confidentiality and will remain anonymous. Your identity will not be disclosed in any part of the study, and the collected data will only be used for academic and research purposes.

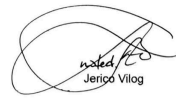
We sincerely hope for your kind consideration and participation in this study. Your insights would greatly contribute to the success and completion of our research.

Thank you very much for your time and support.

Respectfully yours,

Alday, John Denrick M.
Canuel, Mark Justin R.
Eusebio, Rain Justine R.
Familara, Carl Justin B.
Gaoat, Athena Jem D.
Huerto, Benlester N.
Laguetao, Maria Angela B.
Parale, Carl Kristian A.
Singian, Em-Jay P.
Verdan, Geryk Drew P.

Researchers,
Bachelor of Science in Information Technology (BSIT),
ICCT Colleges


Jerico Villog

Appendix C: Pre-Implementation Survey Questionnaire

Purpose: This survey aims to identify the current challenges students face in accessing school document requirement information at ICCT Cainta Main Campus.

Instructions: Please answer the following questions honestly. Your responses will be kept strictly confidential.

Scale:

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

Section A: Information Fragmentation

1. I typically need to check multiple platforms or sources (e.g., Facebook, website, bulletin boards, friends) to find complete information about school requirements or school information.
2. Do you find it difficult to locate important information (e.g., requirements, procedure, announcements) on the organization's Facebook page or Facebook group because it gets mixed or buried under many posts?

3. I encounter conflicting or outdated information about school requirements across different platforms.
4. Searching for a single school document requirement (e.g., good moral, prospectus, transcript of records, etc.) takes me a significant amount of time.
5. The information I usually find about school requirements in school and multiple online platforms is sometimes not clear and confusing.
6. I have failed to complete a school requirement because the instructions I found were unclear or incomplete.

- 1 – Strongly Disagree

Demographic Profile

Year Level:

- ☐ 1st Year
- ☐ 2nd Year
- ☐ 3rd Year
- ☐ 4th Year

Section B: Repetitive Inquiries

7. I find myself asking the same question about school requirements to different people because the information is incomplete or unclear.
8. I have to ask the same question multiple times because I cannot find a clear answer and to be sure.
9. When I ask a question about school requirements, I often experience delays in receiving an answer.
10. I usually rely on asking friends or staff to find answers about school requirements.
11. Clear and accessible information about school requirements would reduce the need for me to ask repeated questions.

Section C: Information Uncertainty

12. I often feel unsure or doubtful about the information and school requirements procedure.
13. I worry that I might miss an important step because the information or procedure is not clearly explained.
14. I have experienced submitting wrong or incomplete requirements because the instructions weren't clear.
15. I often receive different answers from different people, social media, or offices about the same requirement.
16. The process of finding school requirements is not easy because of scattered or fragmented information and misinformation.

Appendix D: Post-Implementation UAT Survey Questionnaire

Purpose: This survey aims to evaluate the effectiveness, usability, and user acceptance of the eGuide ICCT system after deployment at ICCT Cainta Main Campus.

Instructions: After using the eGuide ICCT system, please indicate your level of agreement with each statement below. Your responses will be kept strictly confidential.

Scale:

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree

Section 1: Information Fragmentation

This section measures whether the system successfully centralized information.

1. eGuide ICCT centralizes all school requirement information in one platform.
2. I can easily find all document requirement information without checking multiple sources (Facebook, bulletin boards, etc.).
3. The system eliminates the problem of requirement posts getting buried or lost.
4. All document requirement information in eGuide ICCT is consistent across different pages and sections.
5. Compared to previous methods, eGuide ICCT has significantly reduced information fragmentation.

Section 2: Repetitive Inquiries (Outcome Evaluation)

This section measures whether the system reduced repeated office visits and questions (based on Table 4.3, Q6–Q10).

6. eGuide ICCT provides clear, step-by-step instructions that I can follow without staff assistance.
7. Using eGuide ICCT would reduce the number of times I need to visit an office for requirement questions.
8. The system helps me submit complete document requirements on my first attempt.
9. eGuide ICCT has reduced the need to ask the same question multiple times.
10. The system reduces the number of times I need to ask the same question, allowing me to complete document requirements more efficiently.

Section 3: Information Uncertainty

This section measures whether the system increased trust and reduced confusion.

11. I trust the requirement information in eGuide ICCT is accurate and official.
12. eGuide ICCT has reduced my confusion about how to interpret requirement instructions.

13. I trust eGuide ICCT more than information found on Facebook or from other students.
14. eGuide ICCT has reduced inconsistent submissions caused by students relying on different sources.
15. Overall, eGuide ICCT has significantly reduced information uncertainty and confusion at ICCT.

Section 4: Willingness to Use the System

This section measures adoption and recommendation .

16. eGuide ICCT would be a valuable addition to ICCT's current information dissemination process.
17. Compared to current methods (Facebook, bulletin boards, word-of-mouth), this system is more effective in ensuring students receive accurate requirement information.
18. The system makes it easier to disseminate information and requirement updates to students.
19. I would actively use and recommend eGuide ICCT to other students or administrators at ICCT.

Section 5: Functional Testing

This section tests whether the system functions as intended.

F1. I was able to log in from the system.

F2. I was able to access the system without errors.

F3. I was able to view school document requirements and procedures.

F4. I was able to view step-by-step guides.

F5. I was able to search for specific document requirements procedures.

F6. I was able to see announcements and updates.

F7. I received email notifications when admin posted announcements.

F8. I was able to download/save document requirements procedure as PDF.

F9. The system worked properly on my device.

F10. I was able to search for specific announcements and updates.

F11. The system showed an appropriate message when my search had no results.

F12. I was able to create a new account (sign up) successfully.

F13. I received an OTP (one-time password) when I signed up.

F14. I was able to reset my password using the "Forgot Password" feature.

F15. I received an OTP when I requested to reset my password.

F16. I was able to log in using my new password after resetting it.

Appendix E: Administrative Staff Interview Guide

Purpose: This interview aims to gather qualitative feedback from administrative staff regarding the usability, functionality, and overall effectiveness of the eGuide ICCT system.

Interview Questions

1. How easy or difficult was it to navigate the admin dashboard?
2. How straightforward was the process of posting announcements and updates?
3. How easy was it to add, edit, or delete requirement guides?
4. Overall, how satisfied are you with the admin dashboard?
5. Are there any security concerns you would like to raise?
6. What is your overall impression of the eGuide ICCT system?
7. What features or improvements would you like to see in future versions of the system?
8. Do you have any suggestions for improving the login process?
9. Do you have any other comments or suggestions?